

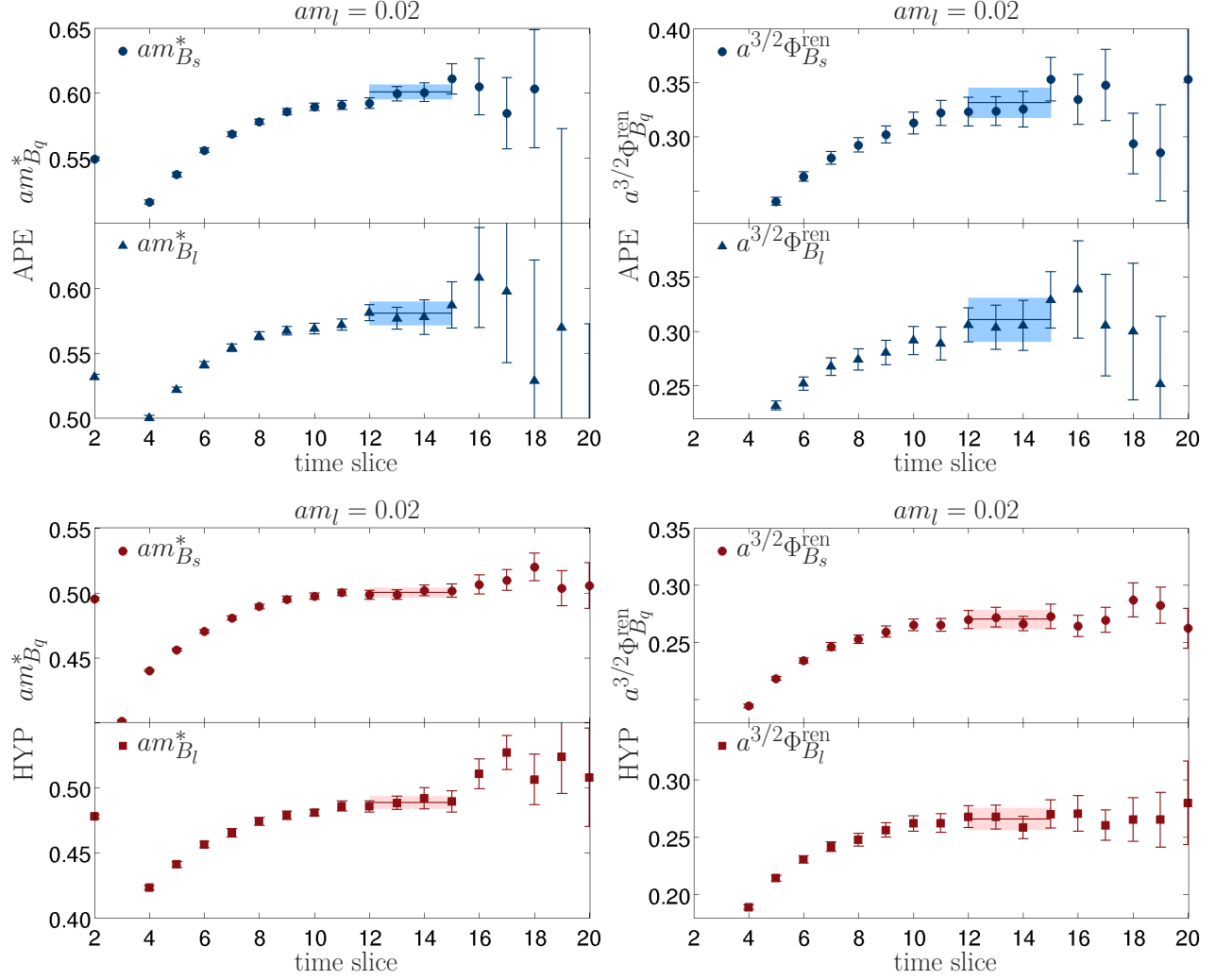


FIG. 1: Determination of $m_{B_q}^*$ (left) and $\Phi_{B_q}^{\text{ren}}$ (right) using the average of two time sources on the $am_l = 0.02$ ensemble. The upper plots show the APE data, while the lower plots show the HYP-smear data. For each panel, the shaded band corresponds to the plateau extracted from averaging the data over four consecutive time slices. Errors shown are statistical only.

matrix element divided by the B_q -meson mass:

$$M_{B_q} = \mathcal{M}_q / m_{B_q}. \quad (66)$$

For the determination of M_{B_q} , we use different spatial sources for the two choices of link smearing: in the case of APE-smear links we use box sources of size 8^3 , while for HYP smearing we use wall sources. The local-wall two-point functions are already defined in